Keywords: legged robotics • reinforcement learning • spatiotemporal control • dynamic interception

Spatiotemporal Agility: Time-Constrained Reinforcement Learning for Vision-Guided Dynamic Quadrupedal Interception

Zhejiang University Team Replaces Velocity-Tracking Policies with Direct Temporal-Spatial Conditioning to Enable Quadruped Ball-Catching Under Strict Deadlines

By Yidong Zhu, Zibo Dai, Tongning Zhang, Leixin Chang, Hua Chen (Zhejiang University; LimX Dynamics)

arXiv:2608.06907 • Preprint, August 2026

A quadruped robot catching a thrown ball must predict where and when the ball lands from visual observations, then reach that position before the deadline. A new paper from Zhejiang University and LimX Dynamics argues that the standard velocity-tracking locomotion paradigm is fundamentally mismatched to this class of tasks, and proposes a unified framework that conditions an RL policy directly on spatial targets and temporal deadlines rather than velocity commands.

AT A GLANCE

Problem: Velocity-tracking quadruped policies cannot meet strict temporal deadlines for reaching precise positions in dynamic interaction tasks.

Why it matters: Dynamic interaction (catching, intercepting, dodging) is the next frontier for legged robot agility; no existing policy architecture handles it.

Method: Vision module predicts landing point and time; RL policy is conditioned on (position, deadline) rather than velocity commands.

Result: Ball-catching benchmark significantly outperforms velocity-tracking decomposition baselines.

Evidence: Evaluated on a standardized ball-catching task with randomized trajectory and timing.

The Big Picture

Legged robots have progressed from laboratory curiosities to practical platforms over the past decade, largely driven by RL-based locomotion policies that excel at navigating diverse terrain. But the benchmark tasks that drove this progress—walking on uneven ground, stair climbing, recovering from pushes—share a common structure: they reward stable navigation with no hard temporal deadline.

Dynamic interaction tasks are qualitatively different. A robot catching a ball, intercepting a rolling object, or dodging a thrown projectile must solve a joint **spatiotemporal problem**: arrive at a specific location by a specific time. Velocity-tracking policies, which accept a desired velocity vector and try to track it, have no native representation of temporal deadlines. Even if they could in principle navigate to any position, they cannot guarantee arrival by a deadline, making them unsuitable for dynamic interception.

Why Existing Approaches Fall Short

The standard decomposition for legged robot tasks is:

- **High-level planner:** computes a path and velocity profile
- **Low-level controller:** tracks the desired velocity command

This decomposition fails for time-constrained interception for two reasons. First, velocity-tracking introduces an abstraction layer that discards the temporal deadline: the planner must guess a velocity profile that achieves arrival by the deadline, but the tracking error of the low-level controller means the actual arrival time is unpredictable.

Second, the two-level decomposition requires the planner to predict the controller's behavior, introducing compounding errors. In practice, systems that work well for slow, navigational tasks become brittle when deadlines are tight and the target is moving.

Core Mechanism

Vision Module. A neural vision module processes monocular camera images to predict:

- $\hat{\mathbf{p}} \in \mathbb{R}^3$: predicted landing position (world frame)
- $\hat{\tau} \in \mathbb{R}_{>0}$: predicted time-to-landing (seconds)

The module is trained on ball-trajectory data with supervised regression.

Time-Conditioned RL Policy. Rather than consuming velocity commands, the locomotion policy is conditioned on the pair $(\hat{\mathbf{p}}, \hat{\tau})$. This directly expresses the spatiotemporal constraint: reach position $\hat{\mathbf{p}}$ within $\hat{\tau}$ seconds. The policy must learn internally to compute appropriate gaits, step frequencies, and postures to meet both the spatial and temporal objectives simultaneously.

Technical Formulation

Let the robot state at time t be s_t and the action (joint torques or velocity targets) be a_t. The policy $\pi_\theta(a_t \mid s_t, \hat{\mathbf{p}}, \hat{\tau} - t)$ is conditioned on the remaining time budget $\hat{\tau} - t$, which decreases with each timestep, providing an implicit urgency signal.

The reward is:

$$r_t = \lambda_{\text{pos}}\, r_{\text{pos}}(s_T, \hat{\mathbf{p}}) + \lambda_{\text{time}}\, r_{\text{time}}(T, \hat{\tau}) + \lambda_{\text{stab}}\, r_{\text{stab}}(s_t)$$

where T is the actual interception time, r_{pos} rewards proximity to the landing point, r_{time} penalizes arriving after the deadline, and r_{stab} encourages stable locomotion.

The vision module and locomotion policy are trained jointly, with the RL reward backpropagating a signal to align vision predictions with achievable interception targets.

Learning or Inference Procedure

Simulation training: The robot and ball physics are simulated with randomized launch parameters (speed, angle, direction) to cover diverse trajectory families. Domain randomization is applied to robot dynamics to improve sim-to-real transfer.

Curriculum: Training begins with slow, predictable ball trajectories and gradually increases speed and variability as the policy improves.

Inference: The vision module processes each incoming frame, updating $\hat{\mathbf{p}}$ and $\hat{\tau}$ in real time. The locomotion policy uses the latest prediction to compute joint targets at 50 Hz.

What the Guarantee Says

No formal guarantee is provided. The policy's behavior near the deadline depends on learned heuristics rather than provable time-optimal control. However, the experimental results demonstrate that the time-conditioned formulation produces significantly more reliable deadline adherence than velocity-tracking baselines, suggesting the implicit urgency signal is effective in practice.

Experimental Findings

- **Ball-catching success rate:** The time-conditioned RL policy significantly outperforms velocity-tracking decomposition baselines on the proposed benchmark.
- **Deadline adherence:** The policy learns to vary gait speed continuously based on remaining time, rather than running at maximum speed and stopping.
- **Trajectory diversity:** Performance degrades gracefully with increasing ball speed and trajectory variability.
- **Sim-to-real:** The framework transfers to physical hardware with domain randomization, demonstrating real-world viability.

Ablations and Interpretation

Time conditioning vs. position-only conditioning. Removing the time dimension and conditioning only on target position degrades success rate substantially, confirming that the temporal deadline is a necessary conditioning input rather than information the policy can infer from position alone.

Joint vs. separate training. Training the vision module and locomotion policy separately (first vision, then RL with fixed vision) produces lower success rates than joint training, suggesting that the locomotion policy's behavior affects what landing predictions are achievable.

Curriculum vs. full distribution. Training without curriculum (full ball-speed distribution from the start) results in slower convergence and lower asymptotic performance, particularly for fast trajectories.

Reference

Yidong Zhu, Zibo Dai, Tongning Zhang, Leixin Chang, Hua Chen. **Spatiotemporal Agility: Time-Constrained Reinforcement Learning for Vision-Guided Dynamic Quadrupedal Interception**. arXiv:2608.06907, August 2026. https://arxiv.org/abs/2608.06907